

MATERNOVA

System Requirements & Prerequisites for API Integration

1. Anthropic Claude API (Core AI Engine)

Authentication	API Key (Bearer Token) passed in headers.
Network Protocol	HTTPS (TLS 1.2 or higher required).
Data Format	JSON payload. <code>application/json</code>
Image Processing Requirements	- Supported Formats: JPEG, PNG, GIF, WebP. - Max File Size: ~5MB (Base64 encoded string within JSON). - Client-side compression (canvas resize) recommended before upload.
Client-Side Prerequisite	Browser support for HTML5 <code><canvas></code> and <code>FileReader</code> API for capturing and encoding OCR images locally.

2. Bhashini API (Vernacular Translation & Voice)

Authentication	API Key / User Token generated via Bhashini Developer Portal (Ulca).
Network Protocol	HTTPS / WSS (WebSocket Secure for real-time streaming).
Audio Input Specs (Speech-to-Text)	- Format: raw PCM, WAV, or WebM (browser native). - Sample Rate: 16kHz recommended. - Channels: Mono.
Client-Side Prerequisite	Browser support for <code>MediaRecorder API</code> and explicitly granted Microphone permissions (HTTPS required).

3. Infermedica API (Medical Triage)

Authentication	`App-Id` and `App-Key` passed in custom request headers.
Data Structure	Strict JSON schema adherence. Requires mapping local UI inputs to Infermedica's specific medical concept IDs (e.g., `s_21` for headache).
State Management	Stateless API. The backend must maintain the current diagnostic state (list of gathered symptoms) and resend the complete state with every new question iteration.
Backend Prerequisite	Node.js/Python intermediate server route required to hide API keys (Do NOT call directly from frontend client).

4. DHIS2 API (GovTech Health Sync)

Authentication	Basic Auth or OAuth2 (Bearer Token) depending on the specific government instance configuration.
Data Format	JSON or XML. Strict adherence to DHIS2 DataElement and Tracker schema structures.
Offline Capability (Client-Side)	• IndexedDB or LocalStorage required to queue failed requests. • Service Worker `SyncManager` API to background-sync queued payloads upon network reconnection.
Idempotency	Payloads must include unique client-generated UUIDs to prevent duplicate entries when retrying failed network calls.